AOUN

عـــون

Intelligent Cancer Patient Monitoring System

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Abstract

Aoun (عون — "*support*" in Arabic) is a fully integrated, locally hosted cancer patient monitoring platform. It bridges the critical gap in oncology care: **the absence of continuous, AI-driven monitoring between hospital visits**. The system combines a **Random Forest ML risk engine** (92% accuracy, 743,728 records), an **OpenCV facial wellness analyzer** grounded in peer-reviewed clinical research (Forte et al., 2021), and a **RAG chatbot powered by local Mistral-7B** — delivered through a Flutter mobile app backed by FastAPI & PostgreSQL. Three dedicated dashboards serve patients, oncologists, and administrators.

Key Metrics

92%	743K	27/28	254	23
ML MODEL ACCURACY	TRAINING RECORDS	FRS IMPLEMENTED	RAG KNOWLEDGE ENTRIES	CLINICAL FEATURES

Problem Statement

1. **Reactive monitoring** — data captured only at clinic appointments, missing daily symptom changes
2. **No AI risk stratification** — no tool to identify high-risk patients before symptoms become critical
3. **No intelligent assistant** — patients lack access to evidence-based answers grounded in their own clinical history
4. **Inefficient triage** — oncologists waste time on low-risk cases instead of flagged high-risk patients

Core Features

FEATURE 01 — ML Risk Prediction Engine

Random Forest classifier trained on 743,728 oncology records predicts High / Medium / Low risk from 23 clinical features. Automated Firebase push alerts dispatched to the oncologist on every High-risk result.

Algorithm: RandomForestClassifier · 100 trees · <200 ms · 92% accuracy
Top feature: Coughing of Blood (12.8% importance)

scikit-learn · FastAPI · Firebase FCM · Port 8001

FEATURE 02 — Facial Wellness Analyzer

OpenCV + Haar cascade pipeline extracts four clinically validated indicators from a patient selfie: pallor (40%), eye fatigue (35%), skin uniformity (15%), skin dullness (10%). Illness score 0–100. No specialist hardware required.

Basis: Forte et al. (2021) Frontiers in Medicine · LAB colour space
Detection rate: 88% on Samsung S918B · EXIF rotation: 100% corrected

OpenCV · Haar Cascade · LAB Colour · CPU-only

FEATURE 03 — RAG Health Chatbot

Retrieval-Augmented Generation chatbot using Mistral-7B (local via Ollama) + ChromaDB vector store indexed on 254 NCI/ACS entries. Feedback-weighted re-ranking improves answers over time. Emergency and medication guardrails block unsafe queries instantly.

LLM: Mistral-7B-Instruct · all-MiniLM-L6-v2 · 384-dim · Over-fetch 3×
Guardrails: 12/12 emergency phrases blocked · RAG relevance: 87%

Mistral-7B · ChromaDB · Ollama · RAG

FEATURE 04 — Emergency Guidance Screen

Dedicated screen for oncological emergencies (febrile neutropenia, severe bleeding, breathing difficulty). Evidence-based guidance with one-tap emergency calling. Intent classifier bypasses the LLM entirely for zero-latency response.

Trigger: Keyword intent classifier · Hardcoded safe response · No LLM
Action: url_launcher tel: URI · One-tap dial to emergency contact

Flutter · Intent Guard · url_launcher

System Architecture

MOBILE APP	Flutter 3.41 / Dart — 13 screens · Patient, Doctor & Admin dashboards · Tested on Samsung S918B	Android · iOS · Chrome
	REST API over HTTP/JSON	
BACKEND API	FastAPI + Uvicorn — Business logic · Auth · Alerts · Reports · SQLAlchemy ORM	Port 8002
	Internal HTTP · localhost	
ML SERVICE	scikit-learn — Random Forest risk prediction · OpenCV facial wellness analyzer	Port 8001
	Ollama API · localhost	
LOCAL LLM	Mistral-7B via Ollama — Instruction-tuned · No cloud · No API key · Fully private	Port 11434
	SQLAlchemy ORM	
DATABASE	PostgreSQL 18 — 8 tables · Patient data · Alerts · Chat feedback · Symptom logs	Port 5432
	Vector similarity search	
VECTOR STORE	ChromaDB — 254 NCI/ACS entries · all-MiniLM-L6-v2 · 384-dim embeddings	Local
	Firebase Cloud → Doctor device	
PUSH ALERTS	Firebase FCM — High-risk push notifications dispatched to treating oncologist's device	Cloud

Evaluation & User Roles

Requirements & Testing

Functional Requirements Coverage	96.4%
Face Detection Rate (50 photos)	88%
RAG Relevance Rate (30 queries)	87%
Emergency Guardrail (12/12)	100%

20 TEST CASES	19 PASSED
27 FRS DONE	+4 GP2 NEW FRS

User Roles

Patient
Register with full medical profile · Log 21 daily symptoms (scale 1–10)
Lifestyle log (sleep, water, exercise, diet) · View AI risk dashboard
Facial wellness selfie scan · Chat with RAG health assistant
Access emergency screen · View recommendations & 30-day reports

Oncologist / Doctor
Patient triage list sorted High → Medium → Low
Receive Firebase push notifications · Confirm / dispute ML diagnosis
Submit ML feedback (correct / incorrect) · View chatbot analytics

Administrator
Approve new patient registrations · Deactivate accounts
Full system oversight dashboard · Account lockout management

Technology Stack

- Flutter 3.41 / Dart
- FastAPI + Uvicorn
- PostgreSQL 18
- scikit-learn · Random Forest
- OpenCV / cv2
- Mistral-7B via Ollama
- ChromaDB
- Firebase Cloud Messaging
- SQLAlchemy ORM
- all-MiniLM-L6-v2
- fl_chart (Flutter)
- pypdf + NCI/ACS PDFs
- Web Speech API
- Git / GitHub

Future Work

- → Bcrypt password hashing migration (30-min task)
- → JWT authentication tokens for all endpoints
- → FR7: Wearable integration via flutter_health plugin
- → Arabic language chatbot support
- → Clinical validation study with hospital partner
- → Model retraining using accumulated doctor feedback
- → Real-time WebRTC video wellness streaming
- → Wider mobile testing (iOS + more Android devices)

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